

(T6)

$$\xi \sim p(x) = \begin{cases} \frac{\theta-1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases} \quad \text{где } \theta > 1$$

 $\exists \bar{x}_n$ а) ОМП

ф.я правдоподоб.

$$L(\theta) = \prod_{i=1}^n p(x_i; \theta) = \frac{(\theta-1)^n}{\left(\prod_{i=1}^n x_i\right)^\theta} \begin{cases} \min_{i=1, n} x_i \geq 1 \end{cases}$$

$$(\ln L)'_{\theta} = \left(n \ln(\theta-1) - \theta \left(\sum_{i=1}^n \ln x_i \right) \right)'_{\theta} =$$

$$= \frac{n}{\theta-1} - \sum_{i=1}^n \ln x_i = 0 \Rightarrow \tilde{\theta} = 1 + \frac{n}{\sum_{i=1}^n \ln x_i} = \underline{\text{ОМП}}$$

$$(\ln L)''_{\theta} = -\frac{n}{(\theta-1)^2} < 0 \Rightarrow \text{это максимум}$$

б) • Результат• $p(x; \theta)$ не зависит от θ на $(1, +\infty)$

$$\cdot \int_1^{+\infty} \frac{\partial}{\partial \theta} p(x; \theta) dx = 0$$

$$/* \quad p'_{\theta} = \left(\frac{\theta-1}{x^\theta} \right)'_{\theta} = \frac{1 + \ln x - \theta \ln x}{x^\theta}$$

$$\Rightarrow \int_1^{+\infty} x^{-\theta} (\ln x (1-\theta) + 1) dx = \frac{x^{-\theta+1}}{-\theta+1} \Big|_1^{+\infty} + \int_1^{+\infty} \ln x dx x^{1-\theta} =$$

$$= \frac{1}{\theta-1} - \int_1^{+\infty} x^{-\theta} dx = \frac{1}{\theta-1} - \frac{1}{\theta-1} = 0 \quad */$$

$$I(\theta) = \int_1^{+\infty} \frac{\theta-1}{x^\theta} \left(\ln(\theta-1) - \theta \ln x \right)'_{\theta}^2 dx =$$

$$= \int_1^{+\infty} \frac{\theta-1}{x^\theta} \left(-\ln x + \frac{1}{\theta-1} \right)^2 dx = \int_1^{+\infty} \frac{\ln^2 x}{x^\theta} (\theta-1) - \frac{2(\theta-1)}{x^\theta} \ln x +$$

$$+ \frac{1}{x^\theta(\theta-1)} dx = \frac{1}{(\theta-1)^2} - 2 \int_1^{+\infty} \ln x \cdot \frac{1}{x^\theta} dx +$$

$$+ \int_1^{+\infty} \frac{(\theta-1) \ln^2 x}{x^\theta} dx =$$

$$= \frac{1}{(\theta-1)^2} + 2 \left(\frac{1}{(1-\theta)^2} \right) - \frac{2}{(1-\theta)^2} = \frac{1}{(1-\theta)^2} \quad (\Rightarrow)$$

$$* \int_1^{+\infty} \ln x d \frac{1}{(1-\theta)x^{\theta-1}} = \frac{\ln x}{(1-\theta)x^{\theta-1}} \Big|_1^{+\infty} - \int_1^{+\infty} \frac{x^{-\theta}}{1-\theta} dx =$$

$$= + \frac{1}{(1-\theta)^2} \quad */$$

$$* \int_1^{+\infty} \frac{(\theta-1) \ln^2 x}{x^\theta} dx = \int_1^{+\infty} \ln^2 x d(-x^{1-\theta}) = -x^{1-\theta} \ln^2 x \Big|_1^{+\infty} +$$

$$+ \int_1^{+\infty} 2 \frac{\ln x}{x} x^{1-\theta} dx = 2 \int_1^{+\infty} \ln x d \frac{x^{1-\theta}}{(1-\theta)} = 2 \ln x \cdot \frac{x^{1-\theta}}{1-\theta} \Big|_1^{+\infty} - 2 \int_1^{+\infty} \frac{x^{-\theta}}{1-\theta} dx =$$

$$= + \frac{2}{(1-\theta)^2} \quad */$$

$$(\Rightarrow) I(\theta) = \frac{1}{(1-\theta)^2} \quad \text{непр на } (1+\infty)$$

• $p(x|\theta)$ donec 2 paz непр дискр. по θ на $(1+\infty)$

$$\bullet \int_1^{+\infty} \frac{\partial^l}{\partial \theta^l} p(x|\theta) dx = 0$$

верно для $l=1$

для $l=2$:

$$p''_{\theta\theta} = \ln x \cdot x^{-\theta} (\theta \ln x - \ln x - 2)$$

$$\int_1^{+\infty} (\ln^2 x \cdot x^{-\theta} (\theta-1) - 2 \ln x \cdot x^{-\theta}) dx = 0 \quad \text{т.к. } \ln 1$$

$\Rightarrow$ ОМП соот; асим несмещ, ас. эфф, ас норм.

По определению медианы: $F(x_{0.5}) = 0.5$

$$\text{т.е. } \int_1^{x_{0.5}} \frac{\theta-1}{x^\theta} dx = -x_{0.5}^{1-\theta} + 1 = 0.5 \Rightarrow x_{0.5} = 2^{\frac{1}{\theta-1}}$$

$$\Rightarrow g(\tilde{\theta}) = \frac{\theta-1}{\sqrt{2}}$$

Имеем
$$\frac{\sqrt{n} (g(\tilde{\theta}) - g(\theta))}{\sqrt{\nabla g(\tilde{\theta})^T I^{-1}(\tilde{\theta}) \nabla g(\tilde{\theta})}} \rightsquigarrow N(0,1)$$

$$\nabla g(\tilde{\theta}) = 2^{\frac{1}{\tilde{\theta}-1}} \ln 2 \cdot \frac{-1}{(\tilde{\theta}-1)^2}$$

$$\frac{\sqrt{n} (g(\tilde{\theta}) - g(\theta))(\tilde{\theta}-1)}{2^{\frac{1}{\tilde{\theta}-1}} \cdot \ln 2} \rightsquigarrow N(0,1)$$

$$P\left(g(\tilde{\theta}) - \frac{1.96 \ln 2 \cdot 2^{\frac{1}{\tilde{\theta}-1}}}{\sqrt{n} (\tilde{\theta}-1)} < x_{0.5} < g(\tilde{\theta}) + \frac{1.96 \ln 2 \cdot 2^{\frac{1}{\tilde{\theta}-1}}}{\sqrt{n} (\tilde{\theta}-1)}\right) = \beta$$

- асим. доверит. интервал (ОМН) для медианы

с) ОМН

$$\frac{\sqrt{n} (\tilde{\theta} - \theta)}{\tilde{\theta} - 1} \rightsquigarrow N(0,1)$$

$$\text{// обозн. } \ln \bar{x} = \sum_{i=1}^n \ln x_i$$

$$-1.96 < \sqrt{n} \ln \bar{x} \left(1 + \frac{1}{\ln \bar{x}} - \theta\right) < 1.96$$

$$P\left(1 - \frac{1.96 - \sqrt{n}}{\sqrt{n} \ln \bar{x}} < \theta < 1 + \frac{1.96 + \sqrt{n}}{\sqrt{n} \ln \bar{x}}\right) = \beta$$

асим. доверит. интервал (ОМН)